Federico Milana

Machine Learning Engineer with a Ph.D. in Human-AI Interaction and 5+ years of experience building and evaluating ML and LLM systems. Specialized in production ML and model evaluation.

EXPERIENCE

Machine Learning Engineer

September 2025 – Present / Cleo

• Distilled GPT-5 transaction enrichment into two LoRA-adapted Llama-3.1-8B models, creating low-cost Oracle Sets for drift detection and retraining.
• Developed and evaluated an encoder-only BERT router for multi-agent orchestration, targeting a 10× latency reduction (500 ms to 50 ms).
• Built evaluation pipelines to measure model quality and compare model and prompt iterations before deployment.
• Developed and shipped multimodal ad labelling pipelines and production models for predicted lifetime value (pLTV).
• Built Python data, training and inference pipelines with Flyte, MLflow, AWS and Docker.
• Collaborated with product and engineering teams to deliver production ML solutions.

Postgraduate Teaching Assistant

January 2021 – May 2024 / University College London

Assisted in teaching Python in practical tutorials and assessed coursework for Human-Computer Interaction MSc students.

Research Assistant

August 2019 – November 2019 / University College London

Evaluated the effects of UI design factors on user trust and autonomy delegation to chatbots, contributing to recommendations for chatbot interfaces.

EDUCATION

Ph.D. in Human-AI Interaction in Machine Learning

September 2020 – November 2024 / University College London

Thesis: “Evaluating Interaction with Machine Learning Text Classifiers and Interpretability Techniques”.
Conducted human-AI interaction research with more than 200 participants, designing an interactive text-classification tool and evaluating LIME, SHAP and LLM-generated explanations.

MSc in Human-Computer Interaction

September 2018 – August 2019 / University College London

Graduated with Distinction, awarded Dean’s List honors for top 5% academic performance.

BSc in Computer Science

September 2015 – August 2018 / King’s College London

Graduated with First-class honors. Specialization in Software Engineering.

PUBLICATIONS

AI-Corpus Translation Studies at Work / Springer Nature / *https://doi.org/10.1007/978-981-95-5933-6*

April 2025 – August 2025 / *https://github.com/fmilana/QLoRaBot*

Applied Generative AI to digital Corpus Translation Studies, streamlining analysis of seventeenth-century translations while ensuring reliability through cross-validation and critical oversight.

Understanding Interaction with Machine Learning through a Thematic Analysis Coding Assistant: A User Study / CSCW ’25 / *https://doi.org/10.1145/3711095*

Explored how an Interactive ML tool facilitates critical reflection, insights, and adaptation in analytical processes while uncovering misconceptions about ML in non-expert users.

Chatbots as Advisers: the Effects of Response Variability and Reply Suggestion Buttons / CUI ’23 / *https://doi.org/10.1145/3571884.3597132*

November 2023 – July 2024 / *https://github.com/fmilana/explanations*

Demonstrated how response variability, human likeness and reply suggestion buttons significantly increase chatbot advice following.

CONTACT

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https://github.com/fmilana

SKILLS

Machine Learning
Open-source and API LLMs
NLP
GenAI
Model Evaluation
Explainability
PyTorch
Hugging Face
scikit-learn
XGBoost/LightGBM

Engineering
Python
SQL
Git
Docker
CI/CD
Amazon Redshift
PostgreSQL
DBT

LLM Engineering
OpenAI API
Anthropic API
LLaMA
QLoRA
Transformers
PEFT
TRL
CUDA
bitsandbytes
vLLM

MLOps & Cloud
AWS
Kubernetes
Terraform
Airflow
Flyte
MLFlow
Arize

CERTIFICATIONS

Deep Learning
February 2025
DeepLearning.AI

Machine Learning
February 2025
Stanford University

Mathematics for ML
January 2025
Imperial College London